
MA 780: Topics on Semigroup Theory

Course Code	MA 780
Course Name	Topics on Semigroup Theory
L-T-P-C	3-0-0-3
Prerequisites	Real Analysis
Intended for	PG
Distribution	Elective
Approval	4th Senate (May 2013)

Course Contents

- **Linear dynamical systems (semigroup approach)**
Cauchy functional equation, finite dimensional system (matrix semigroups), uniformly continuous operator semigroups. [10 Lectures]
- **Semigroup generators and resolvents**
Generator of semigroup and their resolvents, Hille Yosida generalization theorems, special classes of semigroups. [15 Lectures]
- **Spectral theory for semigroups and generators**
Spectral theory for closed operators, spectrum of semigroups and generators. [10 Lectures]
- **Semigroups for population equations.** [7 Lectures]

Text Books

1. V. Arnold, **Ordinary differential equations**, 1973.
2. Engel and Nagel, **One-Parameter Semigroups for Linear Evolution Equations**, Springer, 1999.

References

1. Pazy, **Semigroups of linear operators and applications to partial differential equations**, springer, 1983.
2. S. Kesavan, **Nonlinear functional analysis**, Hindustan, 2004.